

2.2 Integrating using Trigonometric Identities

Review:

In this section we will be making use of some trig. Identities. In particular you should recall:

General Idea:

How could you use one of these identities to find $\int \cos^3(x) \, dx$?

Example 1: Calculate the exact value of $\int_0^{\pi/4} \cos^2(x) \, dx$.

Example 2: Find $\int \sin^4(x) \, dx$.

Strategy for Evaluating $\int \sin^m(x) \cos^n(x) \, dx$:

1.

2.

3.

Example 3 (you try): Find $\int \sin^3(x) \cos^5(x) \, dx$

Example 4: Find $\int \tan^4(x) \sec^4(x) \, dx$

Example 5 (you try): Find $\int \tan^5(\theta) \sec^7(\theta) d\theta$

Strategy for Evaluating $\int \tan^m(x) \sec^n(x) dx$:

1.

2.

Useful formulas:

You should recall that

$$\int \tan(x) dx = -\ln|\cos(x)| + C = \ln|\sec(x)| + C$$

Example 6: Find $\int \sec^3(x) dx$.